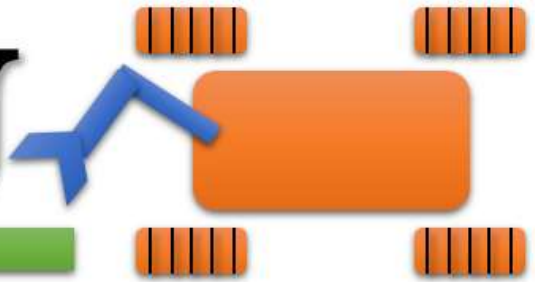


IRoC-U



ISRO Robotics Challenge - URSC

Let's build a space robot

Design Report

by

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10624

**SHRI GURU GOBIND SINGHJI
INSTITUTE OF ENGINEERING &
TECHNOLOGY**

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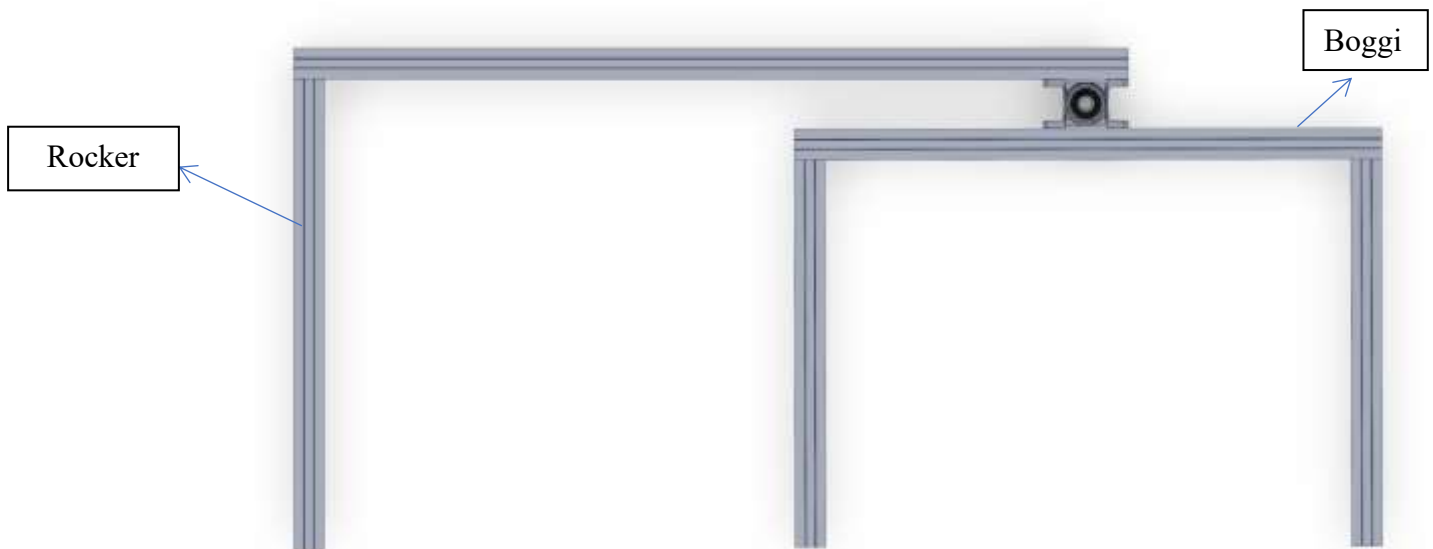
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Rover Mobility

MECHANISM DESIGN

Suspension system:

1. The term “rocker” describes the rocking aspect of the larger links present each side of the suspension system and balance the bogie as these rockers are connected to each other and the vehicle chassis through a modified differential.
2. In the system, “bogie” refers to the conjoining links that have a drive wheel attached at each end
3. As accordance with the motion to maintain centre of gravity of entire vehicle, when one rocker moves upward, the other goes down. The chassis plays vital role to maintain the average pitch angle of both rockers by allowing both rockers to move as per the situation



The rocker-bogie design has no springs or stub axles for each wheel, allowing the rover to climb over obstacles, such as rocks, that are up to twice the wheel's diameter in size while keeping all six wheels on the ground. As with any suspension system, the tilt stability is limited by the height of the centre of gravity. Systems using springs tend to tip more easily as the loaded side yields

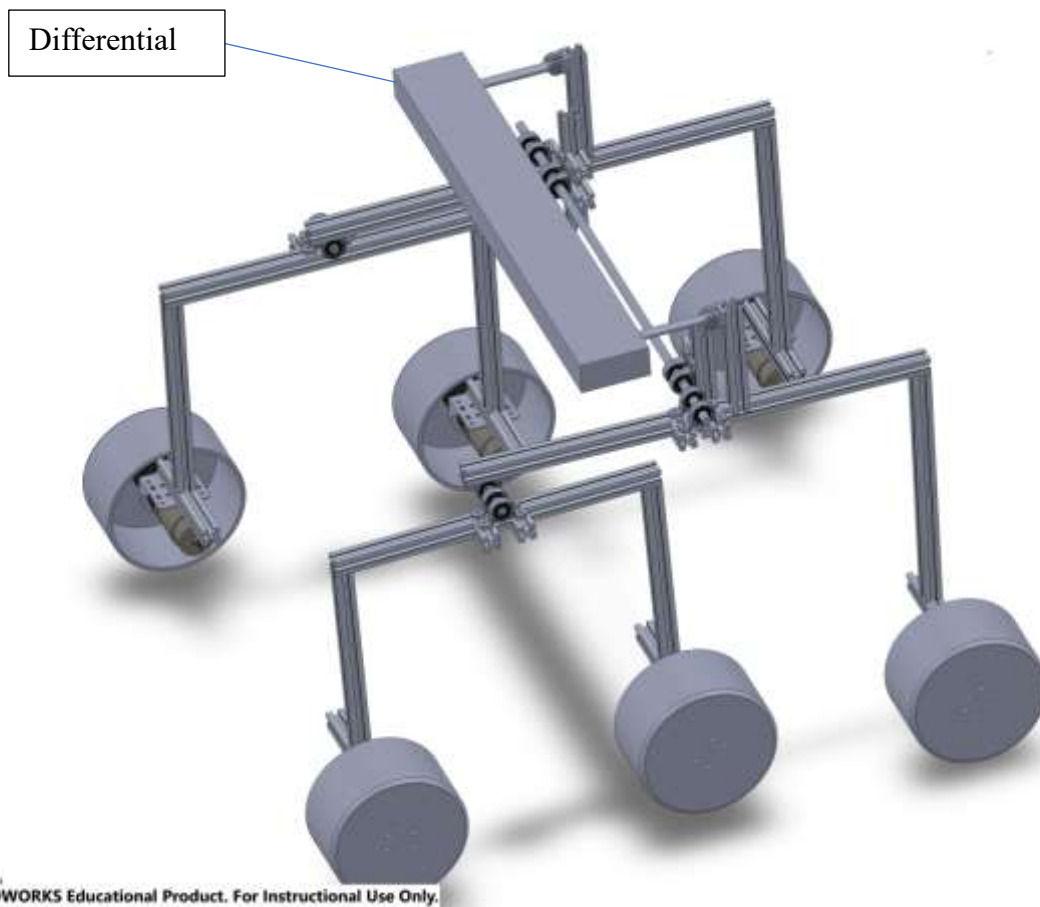
1. The mechanism is designed such as due to the independent motion of right and left rockers, the pitching of the chassis or the rover body remains an average of the two rockers
2. The design incorporates independent motors for each wheel. There are no springs or axles, making the design simpler and more reliable.
3. The design reduces the main body motion by half, compared to any other suspension. The jerk experienced by any of the wheel is transferred to the body as a rotation via the

differential connecting the two rockers, not as translation like conventional suspensions.

4. The rocker-bogie suspension is a mechanism that, along with a differential, enables a six-wheeled vehicle to passively keep all six wheels in contact with a surface even when driving on severely uneven terrain

Differential:

Rover mobility systems play a critical role in enabling exploration missions by facilitating movement across diverse terrains. The rod differential is a fundamental component within rover drivetrains, contributing to enhanced manoeuvrability, traction, and control. This report elucidates the functionality, design, and advantages of rod differentials in rover mobility systems.



1. Overview of Rod Differential:

A rod differential, also known as a torque-biasing differential or limited-slip differential, is a mechanism that allows for the distribution of torque between the wheels of a rover.

Unlike traditional open differentials, which distribute torque equally to all wheels, a rod

differential can vary torque distribution based on wheel slippage and traction conditions.

2. Functionality:

The rod differential operates by utilizing a set of interconnected rods or mechanisms that transfer torque between the wheels.

When one wheel experiences slippage, such as when traversing uneven terrain or encountering obstacles, the rod differential transfers torque to the wheel with greater traction, thereby improving overall traction and stability.

By enabling torque distribution based on wheel conditions, the rod differential enhances rover manoeuvrability and performance across various terrains.

3. Design Considerations:

The design of a rod differential involves careful consideration of factors such as torque bias ratio, material selection, and durability.

The torque bias ratio determines the extent to which torque is transferred from one wheel to another based on slippage conditions.

Materials with high strength-to-weight ratios and resistance to wear are chosen to ensure the reliability and longevity of the rod differential in demanding operational environments.

4. Advantages of Rod Differential:

Improved Traction: The rod differential enhances traction by redirecting torque to wheels with better grip, enabling the rover to traverse challenging terrains with reduced slippage.

Enhanced Manoeuvrability: By dynamically adjusting torque distribution, the rod differential improves the rover's ability to navigate obstacles, negotiate inclines, and maintain stability during turns.

Minimal Impact on Power Consumption: Unlike locking differentials that may increase power consumption, the rod differential operates efficiently by transferring torque only when necessary, optimizing energy utilization during rover operations.

5. Applications and Implementation:

Rod differentials find applications in various rover designs, including planetary exploration rovers and terrestrial vehicles used in rugged environments.

Implementation involves integrating the rod differential into the rover's drivetrain system, ensuring compatibility with other components such as motors, wheels, and control systems.

Testing and validation procedures are conducted to assess the performance and reliability of the rod differential under simulated and real-world operating conditions.

The Rocker Mobility Test is a specific evaluation used in physical therapy and rehabilitation settings to assess an individual's dynamic balance and mobility. It focuses on the ability to shift weight and maintain stability during walking or other movements. Here's an outline of what the test typically involves:

1.Initial Assessment: The individual's baseline mobility and balance are assessed through observation, subjective questioning, and possibly other standardized tests.

2.Test Setup: The individual is asked to stand with feet together or in a comfortable stance on a firm surface.

3.Rocker Board: The individual may be asked to stand on a rocker board or similar device designed to challenge balance. A rocker board typically consists of a platform that pivots in the middle, requiring the individual to maintain balance by shifting weight and adjusting their position.

4.Task Execution: Depending on the specific protocol, the individual may be instructed to perform various tasks on the rocker board, such as:

- Shifting weight from side to side while maintaining balance.
- Rotating the board in different directions.
- Performing reaching or bending movements while standing on the board.

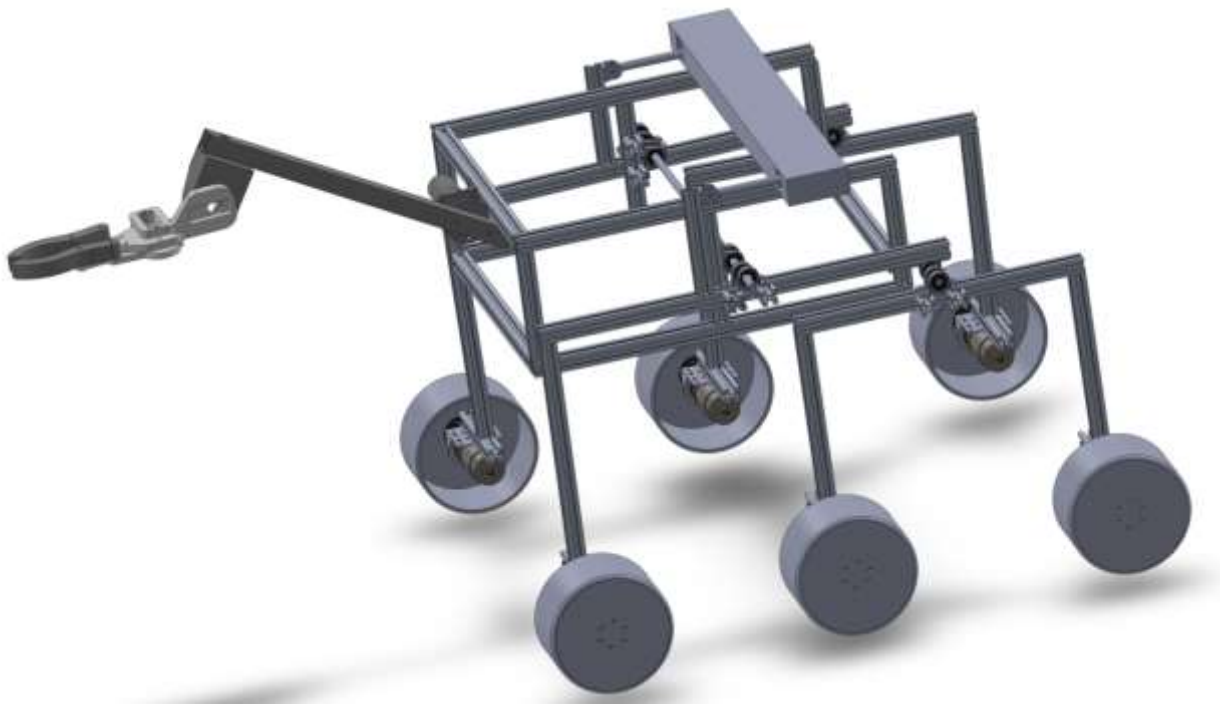
5.Assessment Criteria: The therapist or evaluator observes the individual's performance, looking for signs of instability, loss of balance, compensatory movements, or difficulty completing the tasks.

6.Scoring: The individual's performance may be scored based on predefined criteria, such as the ability to maintain balance, the range of motion during movements, and the quality of movement execution.

7.Interpretation: The results of the Rocker Mobility Test are interpreted in the context of the individual's overall mobility and functional abilities. They may inform treatment planning, goal setting, and interventions aimed at improving balance and mobility.

Overall, the Rocker Mobility Test provides valuable information about an individual's dynamic balance control and can help guide rehabilitation strategies for improving mobility and reducing

the risk of falls or injuries.

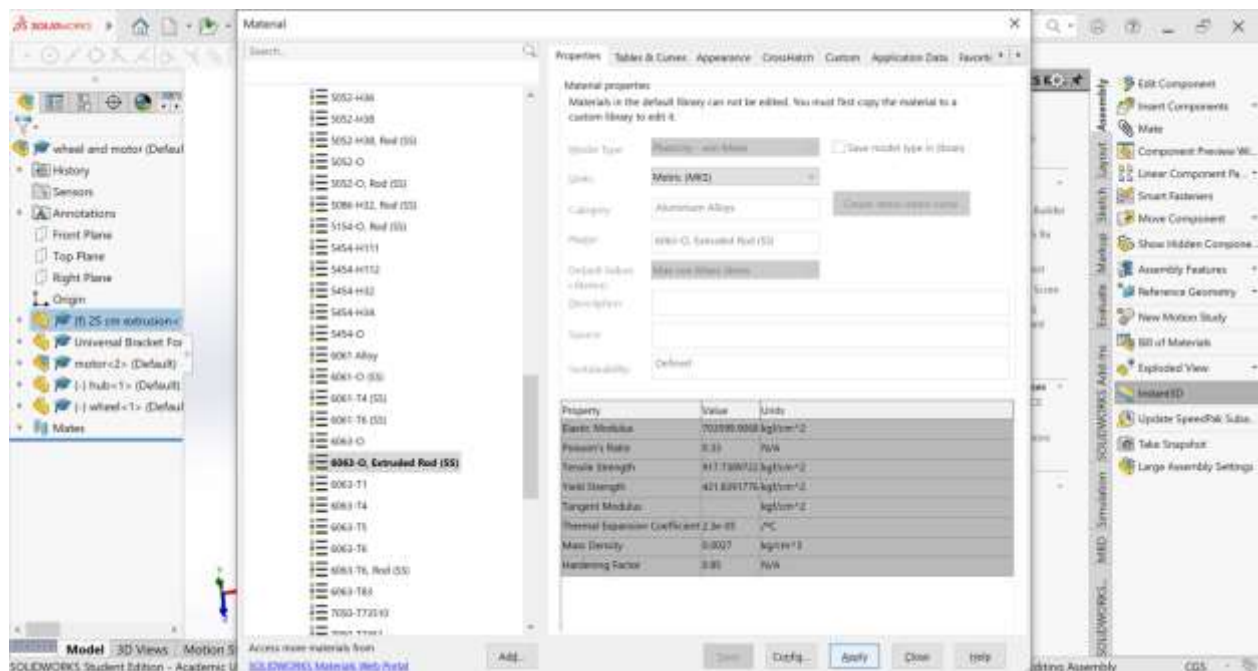


Selection of material is an important step in designing of any component. The main advantages of material selection are:

- It increases the reliability of product
- It reduces the cost of product
- It can also optimize the weight of product

We have selected aluminium alloy that is extrusion which is light weight and it has higher strength among the materials in that weight category.

Following image shows the data of the material test of the extrusion.



Wheels: wheels are the fundamental to the mobility of the rover, dictating its ability to traverse various terrains.

The important things to take in the consideration before designing the wheels are as follow:

1) Terrain compatibility: The ability of rover wheels to navigate diverse terrains encountered during exploration missions is crucial for mission success.

Rover wheels must navigate over rugged and uneven rocky surfaces commonly found on planetary bodies like Mars or the Moon.

Increased wheel diameter and width can help traverse larger obstacles and maintain stability over rocky terrain.

The wheels can encounter the various degree slopes.

2) Material composition: Rover wheels are typically constructed from lightweight yet durable materials to withstand the rigors of extraterrestrial exploration.

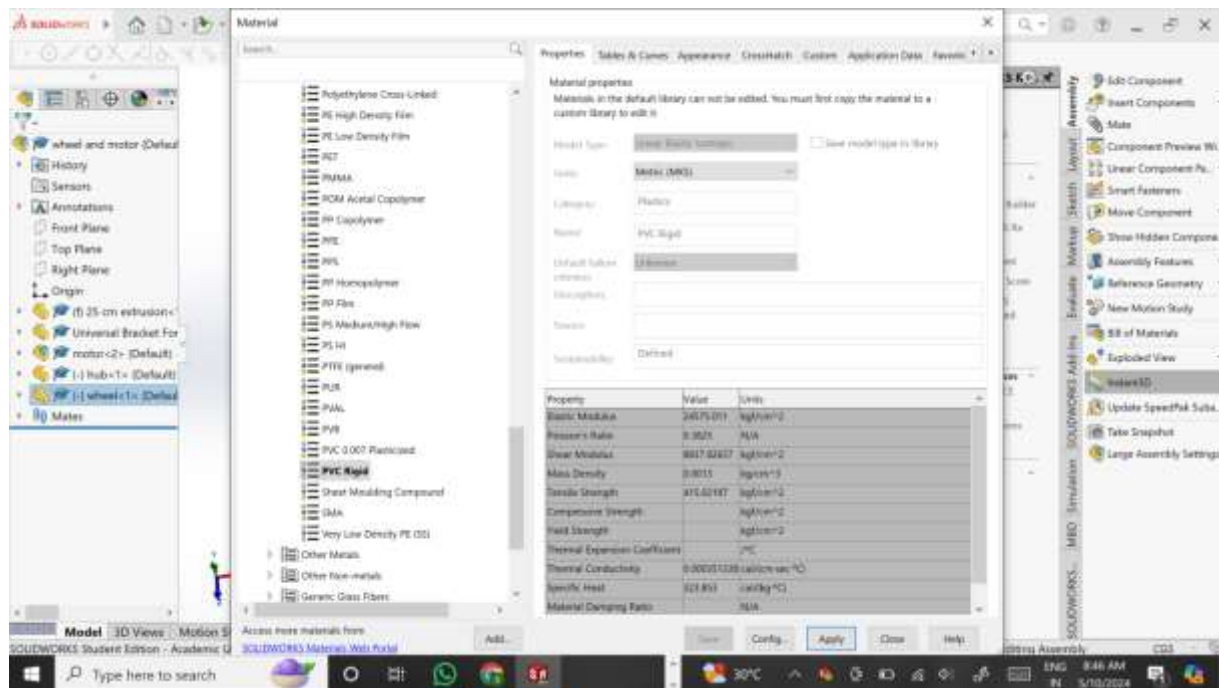
Keeping this aspect in mind we have choose a lightweight and extra durable material which is light weight too that material is polyvinyl chloride.

3) Wheels Flexibility: Flexibility in wheel design is crucial for traversing uneven terrain and absorbing shocks to minimize impact on the rover's chassis and internal components.

4) Dimensions: The diameter of the is 18cm and the wheels are hollow.

5) Cost Efficient: As the wheels are made up of the polyvinyl chloride polymer which is cheap as compared to the other materials having the same specifications and the wheels are

designed such that the internally, they are hollow all the load is evenly distributed over the surface is the key aspect of the success of this type of the wheel.



Wheel

DRIVE ELECTRONICS DESIGN

Motor Selection:

Choose motors suitable for driving the wheels of the rover. These motors should provide sufficient torque and speed control for navigating various terrains. We have chosen the motor such that it can cover 1cm/sec to 5cm/sec. it has the torque of 294.3N/cm with 80rpm.

Motor Controllers:

Utilize motor controllers to regulate the speed and direction of each motor independently. This allows for precise control of the rover's movement, including turning and traversing uneven terrain. According to our requirements, the MDD10A was sufficient to control the motors

Power Supply:

Implement a power supply system capable of providing the necessary voltage and current to drive the motors and other electronic components onboard the rover. This includes batteries with appropriate voltage regulators.

Encoder and IMU Sensor:

Integrate sensors such as encoders, gyroscopes (MPU6050), and accelerometers to provide feedback on the rover's orientation, speed, and terrain conditions. This information is crucial for autonomous navigation and obstacle avoidance.

Microcontroller:

Use a Raspberry Pi5 and ESP32 to process sensor data, execute control algorithms, and communicate with external devices or a central control station.

SENSOR SYSTEM

1. VL53L0X Time-of-Flight (ToF) Sensor

The VL53L0X is a miniature ToF sensor module manufactured by STMicroelectronics. It operates on the principle of emitting laser pulses and measuring the time it takes for the light to return after being reflected by an object, thereby determining the distance to the object. Key features of the VL53L0X sensor include:

- **Distance Measurement Range:** The sensor is capable of accurately measuring distances in the range of up to 2 meters.
- **High Precision:** With its high-resolution measurement capabilities, the VL53L0X sensor provides accurate distance measurements with millimeter-level precision.
- **Compact Size:** The compact form factor of the sensor module makes it suitable for integration into small space
- **I2C Interface:** The sensor communicates with the microcontroller using the I2C (Inter-Integrated Circuit) protocol, allowing for easy interfacing with a wide range of microcontroller platforms.

The VL53L0X sensor is utilized as a key component for distance sensing and obstacle detection. By integrating the sensor with a servo mechanism controlled by an ESP32 microcontroller, the system is capable of performing 2D LiDAR-like scanning, providing real-time distance measurements across a wide field of view.

2. Servo Motor

A servo motor is a rotary actuator that allows for precise control of angular position. a servo motor is employed to enable the VL53L0X sensor to scan across a horizontal plane, expanding the field of view for distance sensing. The servo motor is controlled by the ESP32 microcontroller, which determines the desired angular position based on the scanning pattern requirements.

The servo motor enhances the versatility of the distance sensing system by enabling dynamic adjustment of the scanning angle, allowing for comprehensive coverage of the surrounding environment. By integrating the servo motor with the VL53L0X sensor, the rover achieves enhanced spatial awareness.

3. MPU6050 Inertial Measurement Unit (IMU)

The MPU6050 is a 6-degree-of-freedom (DOF) Inertial Measurement Unit (IMU)

sensor manufactured by InvenSense (now part of TDK). It combines a 3-axis gyroscope and a 3-axis accelerometer in a single package, allowing for the measurement of angular velocity and acceleration along three orthogonal axes. Key features of the MPU6050 IMU include:

- Gyroscope: The gyroscope measures angular velocity, providing information about the rate of rotation around each axis.
- Accelerometer: The accelerometer measures linear acceleration along each axis, enabling the detection of changes in velocity and orientation.
- Digital Interface: The sensor communicates with the microcontroller using the I2C protocol, facilitating easy integration into embedded systems.

The MPU6050 IMU is utilized for orientation estimation and motion tracking. By fusing the data from the gyroscope and accelerometer, the IMU provides real-time updates on the rover orientation and movement

SOFTWARE REQUIREMENT AND DESIGN DETAILS

Overview:

Our project aims to develop an autonomous rover capable of mapping an arena, identifying and locating a pipe within it, performing a pick-and-place operation, and then depositing the pipe at a designated location. This ambitious goal is achieved through the development of a robotic system utilizing ROS2 and various associated components for navigation and pick-and-place tasks. The bot integrates sensor data from an ESP32 microcontroller using Micro-ROS, with the ROS2 framework running on a laptop to process and control the bot's movements and actions. Also we did the simulation of a robotic system using Gazebo, URDF modelling, and the integration of ROS2 components such as Nav2 for navigation and MoveIt for pick-and-place operations. The simulation environment replicates the physical setup and allows for thorough testing and validation of the bot's functionality. Visualization and analysis tools like RViz and rqt were employed to monitor and analyse the system's performance.

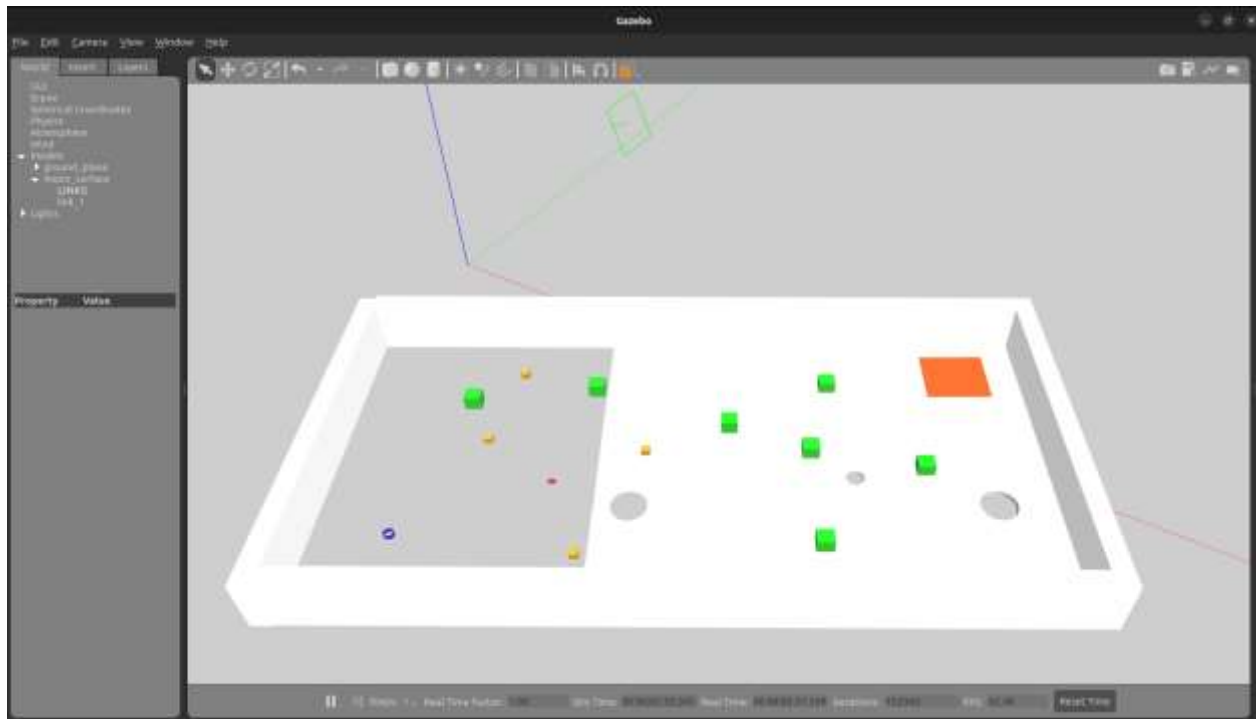
Robot Operating System (ROS2):

We have decided to use ROS2 for our framework due to its robust capabilities in creating and managing nodes and topics, facilitating efficient communication between them. This enables us to visualize and execute the necessary actions to control the rover's motion autonomously. Utilizing frameworks like Nav2 for navigation and SLAM for mapping with a Lidar sensor, the rover will be able to navigate the arena and determine its path independently. Additionally, Nav2 uses the Dynamic Window Approach (DWA) planner to make dynamic decisions. For pick-and-place operations, we use MoveIt to control the robotic arm, allowing it to move to specific coordinates as visualized in RViz, enabling it to grab and place objects as commanded. We can also visualize the bot in RViz and simulate it in Gazebo, both of which are integrated with the ROS2 framework.

Utilisation of hardware:

1. ESP32 Microcontroller:
 - Publishes sensor data on topics /imu, /odom, /camera, and /scan.
 - Subscribes to /cmd_vel and /servo topics to control motor and servo actions.
2. Laptop:
 - Runs the ROS2 framework and processes incoming sensor data.

- Executes Nav2 for navigation tasks.
- Uses MoveIt for pick-and-place operations.



Physical Environment Created on Gazebo

Communication:

- Micro-ROS: Enables communication between the ESP32 and ROS2 on the laptop via UDP.
- UDP Protocol: Facilitates the transmission of sensor data from ESP32 to the laptop and command data from the laptop to the ESP32.

ROS2 Topics:

- /imu: Inertial Measurement Unit data.
- /odom: Odometry data.
- /camera: Camera data for object recognition and manipulation.
- /scan: Lidar or other scanning data for environment mapping.
- /cmd_vel: Command velocities for motor control.
- /servo: Commands for servo control for claw.

Sensor Data Handling:

1. IMU Data (/imu): Provides orientation and acceleration data to aid in navigation and balance.
2. Odometry Data (/odom): Tracks the bot's position and velocity, crucial for path planning and navigation.
3. Camera Data (/camera): Captures images or video for object detection and manipulation.
4. Scan Data (/scan): Provides distance measurements to map the environment and avoid obstacles.

Topic Monitoring:

- rqt: Provided detailed insights into the data being published on various topics such as /imu, /odom, /scan, /camera, /cmd_vel, and /servo. This helped in validating the correctness and performance of the sensor integrations and control commands.

System Visualization:

- RViz: Allowed for real-time monitoring of the bot's navigation, object detection, and manipulation tasks. This visualization ensured that the bot's simulated behavior matched the expected outcomes.

Node and Topic Graph:

- rqt_graph: Displayed the interactions between ROS2 nodes and topics, verifying the proper setup and communication flow within the system. This graph was crucial for debugging and ensuring all components were correctly integrated.

SPECIFICATIONS OF COTS COMPONENTS

COTS Name	Functionalities
Aluminium alloy	Structural material for durability
PVC wheels	Provides mobility and traction
Planetary motor	Drives movement with precision
Battery	Powers the device
Kill switch	Emergency shutdown mechanism
Motor driver	Controls and regulates motor functionality
ESP32	Microcontroller for processing and connectivity
Camera module	Captures visual data
LIDAR	Measures distance with laser technology

Navigation and identification

ALGORITHMS

1. URDF Modelling:

- Fusion 360 Design: The bot was initially designed in Fusion 360, where its physical structure and components were modelled.
- URDF File Creation: The 3D model was exported and converted into a Unified Robot Description Format (URDF) file. This file defines the robot's joints and links, essential for simulating the bot's kinematics and dynamics.

2. Model Creation and Integration:

- 3D Models: Custom 3D models of obstacles and objects were created using tools like Blender or imported from existing Gazebo model libraries.
- Physical Properties: Each model was assigned realistic physical properties such as mass, friction, and collision boundaries to accurately simulate interactions.
- SDF (Simulation Description Format): All objects and structures were defined in a world file using SDF. This file includes the position, orientation, and physical attributes of each object.
- Environment Layout: The world file was carefully designed to include various obstacles, structures, and pathways. This setup allows for comprehensive navigation tests and pick-and-place tasks.

3. Integration with Sensors and Plugins:

- Lidar: Integrated Lidar sensors using Gazebo plugin `gazebo_ros_laser`, is added to the robot model and is configured to publish scan data on a ROS2 topic, `/scan`.
- Camera: A camera sensor plugin, `gazebo_ros_camera`, is attached to the robot's camera model and configured to publish camera images on a ROS2 topic, `/camera`.
- IMU and Odometry: For imu and odometry simulation plugins like `gazebo_ros_imu`, `gazebo_ros_odometry` is added and configured to publish data on ROS2 topics `/imu` and `/odom`, respectively.

SOFTWARE REQUIREMENTS AND ITS DESIGN DETAILS

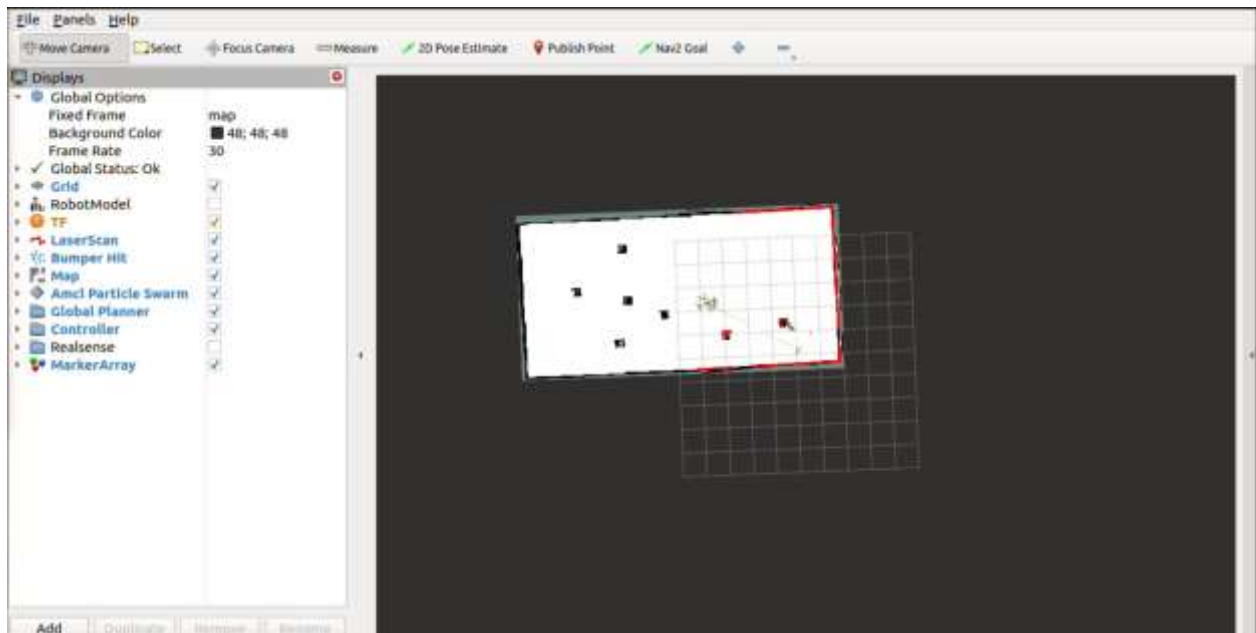
Navigation (Nav2):

Navigation 2, is used for autonomous navigation of the bot.

1. Localization: Utilizes /odom and /imu data to determine the bot's current position within a known map.
2. Mapping: Employs /scan data to update and refine the map of the environment.
3. Path Planning: Generates a path from the bot's current location to a target destination using the updated map.
4. Motion Control: Converts the planned path into velocity commands published on the /cmd_vel topic, which the ESP32 subscribes to and translates into motor actions.

SLAM (Simultaneous Localization and Mapping):

1. SLAM utilizes data from a Lidar sensor that continuously scans the environment, measuring distances to nearby objects and obstacles.
2. SLAM algorithms extract distinctive features from the sensor data, such as corners, edges, or key points. The algorithm identifies and matches locations between different sensor scans.
3. Using the extracted features and data from the robot's motion (odometry), SLAM estimates the robot's pose (position and orientation) within the map.
4. As the robot moves through the environment, SLAM simultaneously constructs a map of the surroundings.
5. It incorporates new sensor data into the map, updating the positions of landmarks and obstacles based on the robot's estimated pose.
6. SLAM algorithms identify loop closures, where the robot revisits a previously visited location.



Map Generated Using SLAM

Pick-and-Place Operations (MoveIt):

MoveIt is used for the bot's pick-and-place tasks. It processes the data as follows:

1. Perception: Utilizes the /camera topic to detect and locate objects within the bot's workspace.
2. Planning: Generates a motion plan to move the robotic arm from its current position to the object, then to the place position.
3. Control: Publishes the necessary servo commands on the /servo topic to execute the planned movements. The ESP32 subscribes to this topic and actuates the servos accordingly.

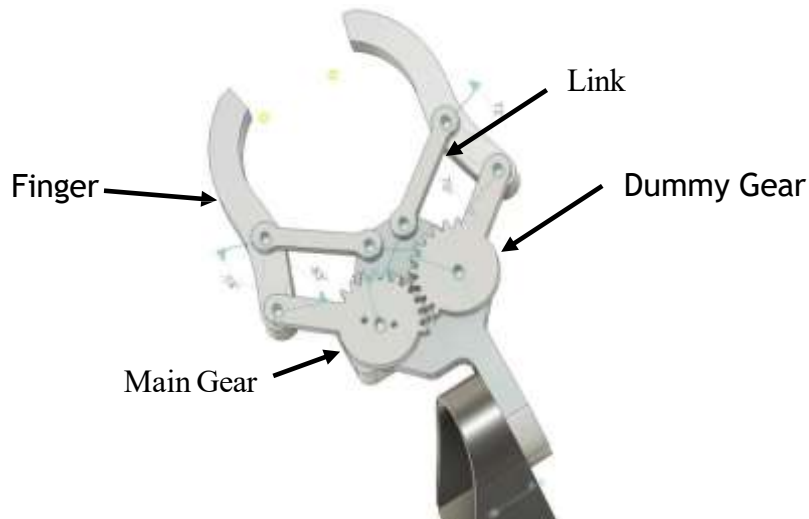
Manipulator

Robotic Arm

- We used two axis robotic arm to pick a place cylinder, which is present on arena.
- Joints plays an important role in movement of gripper and arm, all the joints offer 0°-180° of movement.
- We used planetary and mg995 servo motor for base joint and middle joint respectively.

Gripper

- Gripper used to pick the cylinder from the arena and place it to appropriate position.
- It is operated by mg995 servo motor.
- It consists of main gear and other is a dummy gear, main gear is connected with the servo motor and provide power to its own finger and dummy gear.
- Dummy gear receives the torque from the main gear.
- Direct meshing is used in between two to transmit power.
- We use PLA white filament to make gripper parts.



Justification of used of Direct Meshing Mechanisms

Direct meshing mechanisms are commonly used in grippers for several reasons:

Simplicity: Direct meshing mechanisms are simple in design, typically consisting of just two gears that interlock. This simplicity makes them easy to manufacture, install, and maintain.

Efficiency: Direct meshing mechanisms offer high efficiency because there are minimal frictional losses compared to other mechanisms like belts or chains.

This is important in grippers where precise control and energy efficiency are desired.

Load Handling: Direct meshing systems are effective at handling large loads. In order to provide dependable functioning and minimize premature wear, the gears' teeth distribute the weight evenly, which is crucial for firmly grasping items.

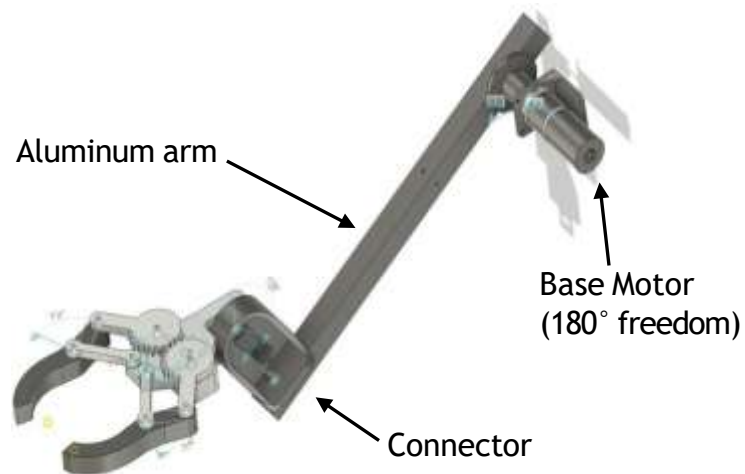
Arm

- A chain of links that are moved by joints which are actuated by motor.
- We use mg995 servo motor at the one end of arm which is near to gripper and planetary motor at the base of arm.
- Both the motors have 180° of freedom.
- A 180cm long hollow aluminium bar is used as arm.

Justification of used Mechanism:

A two-axis robotic arm on a rover can give various benefits and explanations for its use:

- **Simplicity and Reliability:** A two-axis robotic arm is simpler to construct than higher-degree-of-freedom arms, hence it is intrinsically more reliable. There are fewer moving components and joints, reducing the possibility of mechanical failure, which is critical for missions in extreme environments such as space or planetary surfaces.
- **Cost-effectiveness:** A two-axis robotic arm's basic design typically results in reduced production and maintenance costs than more sophisticated arms. This can be beneficial for missions with limited budgets, allowing resources to be directed toward other mission-critical components or objectives.
- **Weight and Power Efficiency:** A two-axis arm often weighs less and takes less power to operate than an arm with more degrees of freedom. This is useful for rovers, where weight and power limits are important in mission design. A lighter arm allows the rover to carry more scientific instruments or cargo, which improves its overall capabilities.

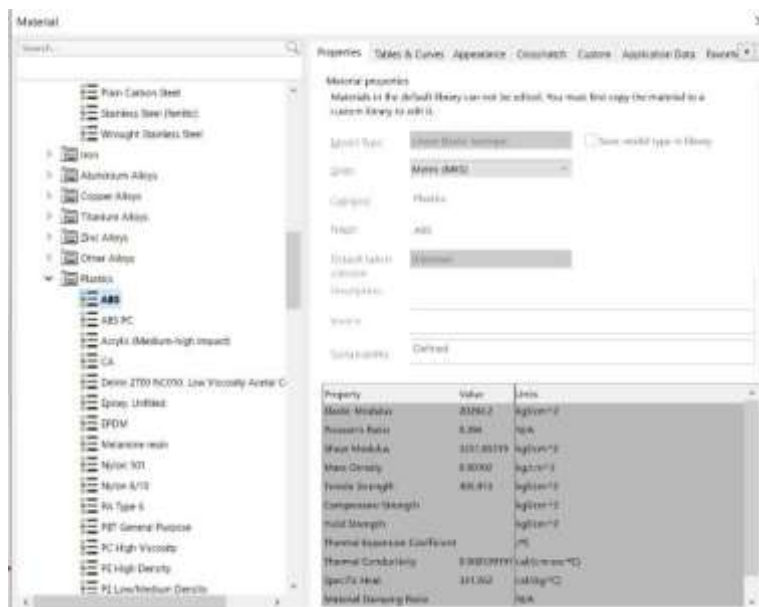


Material used for making of robotic arm.

Selection of material is an important step in designing of any component. The main advantages of material selection are:

- It increases the reliability of product.
- It reduces the cost of product.
- It can also optimize the weight of product.

We have selected aluminum alloy that is extrusion which is light weight and it has higher strength among the materials in that weight category. Following image shows the data of the material test of the extrusion.



CALCULATIONS

From 3D modeling in Solid Works, we got the Centre of mass of Claw = $X_c = 32$ cm,
(Distance X_c is from the horizontal axis of rotation).

Mass of claw (M_c) = 0.8 kg, Mass of pipe (M_r) = 0.2 kg,

COM of the pipe from the horizontal axis of rotation = $X_r = 36.8$ cm.

So, The COM of the whole system (claw + pipe) while picking is X_s .

$$X_s = M_c \times X_c$$

$$M_c + M_r = 0.8 \times 32$$

$$= 25.6$$

$$= 25.6 \text{ cm}$$

$$\text{Torque (T q)} = M_s \times g \times X_s = 2.5 \times 9.81 \times 21.35 = 523.60 \text{ Ncm} = 5.2267 \text{ Nm.}$$

So, we are using a planetary geared motor of 100 rpm and 294.3 N-cm torque to rotate the claw

Calculations for Maximum RPM and Torque of Rover Motors

1. Circumference of the Wheel (C):

- Formula: Circumference (C) = $2 \times \pi \times \text{radius (r)}$
- Radius of the wheel (r): 8 cm (0.08 m)
- Result: Circumference C $\approx$ 0.5027 meters

2. Maximum RPM (R):

- Formula: Maximum RPM (R) = $(V / C) \times 60$
- Maximum speed of the rover (V): 50 cm/s (0.5 m/s)
- Result: Maximum RPM (R) $\approx$ 59.52 RPM

3. Force Required (F):

- Formula: Force (F) = mass (m) $\times$ acceleration (a)
- Mass of the rover (m): 50 kg
- Acceleration of the rover (a): 0.5 m/s^2

4. Torque (T):

- Formula: Torque (T) = Force (F) $\times$ radius (r)
- Force (F) $\approx$ 25 N
- Result: Torque (T) $\approx$ 2 N.m

Communication

RF COMMUNICATION

In our rover design, we utilize two primary communication systems to ensure comprehensive data and video transmission between the rover and the base station. The ESP32 DevKit V4 microcontroller, equipped with a 2.4GHz router antenna, handles sensor data transmission over a distance of up to 25 meters. Additionally, a Raspberry Pi paired with an RF antenna facilitates real-time video transmission to the base station. This dual communication setup ensures reliable data and video transfer, enabling efficient remote monitoring and control of the rover. The design leverages robust wireless capabilities to optimize signal strength and minimize interference, maintaining stable connections in varied environmental conditions.

Component Selection:

Microcontroller: ESP32 DevKit V4

Antenna: 2.4GHz router antenna

Frequency Band: 2.4GHz ISM band

Distance: 25 meters

Key Considerations:

Transmit Power: The ESP32 DevKit V4 typically operates with a transmit power of 20 dBm, ensuring adequate signal strength for the required distance.

Receiver Sensitivity: The receiver sensitivity of the ESP32 is -96 dBm, enabling it to detect weak signals over the specified range.

Antenna Gain: Using 2 dBi gain antennas for both the transmitter and receiver enhances the effective signal strength and coverage.

Performance Optimization:

Signal Margin: The setup provides a sufficient signal margin to ensure reliable communication, accounting for potential obstacles and interference.

Environmental Factors: The design considers potential environmental variations, optimizing placement and orientation of the antennas to maintain a stable connection.

Video Transmission:

Performance:

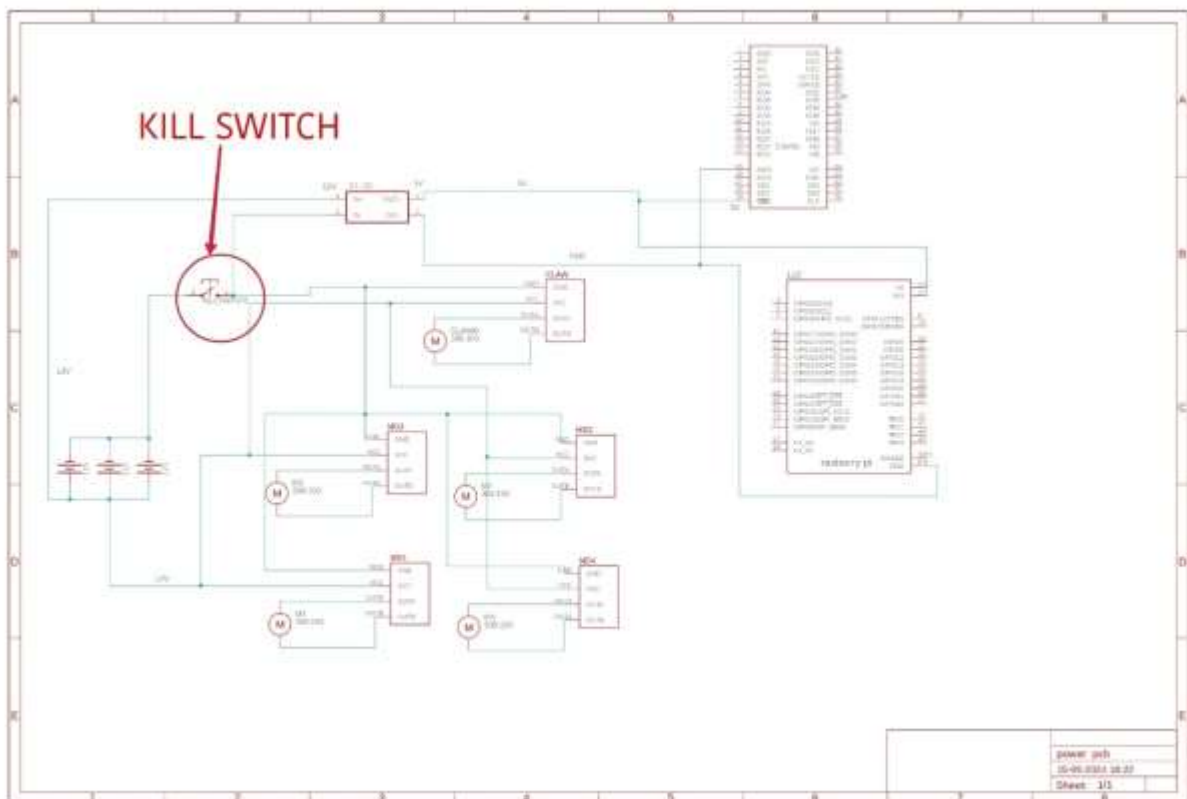
Bandwidth Requirements: Video transmission requires higher bandwidth; thus, selecting an appropriate frequency band and ensuring minimal interference is crucial.

Signal Integrity: Implementing error correction and robust encoding techniques to maintain video quality despite potential signal degradation.

Latency: Minimizing latency is critical for real-time video monitoring, necessitating efficient data handling and transmission protocol

Emergency Response System

In the emergency response system, we have used the kill switch which is a safety mechanism designed to shut down or disable a rover. It is a mechanism which immediately stop an operation by



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Analysis

1. ROVER MECHANISM :

1.Stress Analysis:

- The rocker bogie mechanism is segmented into multiple bogie assemblies connected in series, each comprising two rocker arms and a suspension system for each wheel. Segmentation allows for flexibility and adaptability to uneven terrain by enabling independent movement of each bogie assembly.
- Pivot point techniques are employed at the connection points between rocker arms and the rover chassis to distribute loads evenly and minimize stress concentrations. These techniques ensure that the rover maintains stability and structural integrity even under dynamic loading conditions.

2.Vibration Control with Differential Gear System:

- A differential gear system is integrated into the central portion of the rover to control vibrations generated during movement over uneven terrain. The differential system allows for independent rotation of each wheel while maintaining overall stability and control.
- By distributing torque evenly between wheels, the differential gear system minimizes torsional vibrations and reduces the transmission of vibrations to the rover chassis.
- Vibration analysis is conducted to optimize the design of the differential gear system, ensuring that it effectively dampens vibrations and enhances the rover's stability and performance during traversal of rough terrain.

2.MANIPULATOR MECHANISM:

1.Range of Motion:

The manipulator's range of motion is attributed to its three degrees of freedom (DOF). These degrees of freedom enable the arm to move in various directions within its operational envelope. Each degree of freedom corresponds to a specific joint, allowing for independent movement and positioning of the arm segments. The range of motion encompasses reaching, lifting, rotating, and

tilting motions, providing the manipulator with the versatility needed to accomplish a diverse range of mission tasks. By utilizing its three degrees of freedom effectively, the manipulator can navigate obstacles, access confined spaces, and manipulate objects with precision and agility, enhancing the rover's capabilities for exploration and scientific investigation.

2. Claw Mechanism with Two-Way Gear System:

- At the end-effector of the manipulator arm is a claw mechanism engineered with a two-way gear system, enhancing its functionality for object manipulation. The claw mechanism incorporates a pair of gears that enable both opening and closing motions of the claw's jaws. This two-way gear system provides greater control and flexibility in adjusting the opening width of the claw's mouth, allowing it to accommodate objects of varying sizes and shapes.
- By employing the two-way gear system, the manipulator arm can adjust the grip force exerted by the claw, ensuring secure and stable grasping of objects during manipulation tasks. The gear system enables precise control over the claw's movements, facilitating delicate operations such as picking up small samples or deploying instruments with accuracy.
- Additionally, the two-way gear system enhances the manipulator's adaptability to different mission scenarios and environmental conditions. The ability to adjust the opening width of the claw's mouth ensures efficient handling of objects encountered during exploration, research, and scientific experiments. Overall, the integration of the two-way gear system into the claw mechanism enhances the manipulator arm's capabilities, contributing to the success of the rover mission.

Test Plan

1. Introduction:

Overview: A concise outline of the automated rover project emphasizing its autonomous capabilities and communication functionalities.

Objectives: Ensure robust functionality, reliability, and safety of the rover.

Scope: The test plan covers testing procedures for both hardware and software components under various scenarios.

2. Test Objectives: Verify sensor accuracy and reliability for obstacle detection and avoidance. Ensure seamless communication between the rover and the control station. Assess emergency response mechanisms, including kill switch functionality.

3. Test Environment: Dynamic environments including indoor and outdoor terrains with varied obstacles. Utilize simulated environments for controlled testing of specific scenarios. Access to required equipment such as obstacles, communication devices, and safety measures.

4. Test Components: Hardware: Rocker bogie mechanism, motors, sensors (lidar, MPU6050), MDD10A electronics, batteries, kill switch. Utilize iterative testing methods to refine rover performance. Incorporate feedback loops for continuous improvement.

5. Test Cases: Detailed test cases outlining inputs, expected outputs, and success criteria for each scenario. Flexibility to modify test cases based on evolving project needs.

6. Risk Management: Proactive identification and mitigation of risks associated with testing, ensuring the safety of personnel and equipment.

7. Test Schedule: Agile test scheduling with short iterations, allowing for rapid feedback and adjustment. Prioritization of critical test scenarios based on project milestones and objectives. Regular updates to stakeholders, highlighting progress and areas for improvement.

8. Approval: Efficient approval process with clear criteria for sign-off on test plans and results. Encourage transparency and collaboration among mentors and industrial mentors.

9. References: Include relevant standards, guidelines, and best practices for automated rover testing. Continuously update references to incorporate the latest advancements.

Weighing and dimensions details of a rover

Length	80cm/0.08m
Width	80cm/0.08m
Height	53cm/0.53m
Weight	29.3kg

Link of CAD

https://drive.google.com/file/d/1RtiO5susnDNfdPh_0RzKfM6JwmNXVYwf/view?usp=drive_link